

Database *Carbon substitution in the wood sector*

Data Extraction Protocol

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1 Background

The goal of this data extraction activity is to build a database of wood carbon **balance** values from published sources compiled in a systematic literature search. We will record **pairs of variables** describing the **change in carbon sequestration** and/or emission as a result to a change in wood harvest or use. These carbon budget comparisons will always be linked to **one main driver** judged responsible for the recorded change in wood use i.e. a change in a forest sector practice such as silviculture, supply chain, technology.

The core data of each pair will be the values of the carbon balance and the amount of wood harvested for the two scenarios, a **'reference'** and **'alternative'** driver configurations leading to a reference (Qwoodref , Cref) and alternative (Qwoodalt , Calt) carbon sequestration (Fig. 1). Each carbon budget comparison will have several associated metadata, characterizing the information needed for the analysis and for transparency of the database. All the data and metadata will be recorded in a data template implemented in a spreadsheet. Each data contributor will be assigned a number of papers from the literature search.

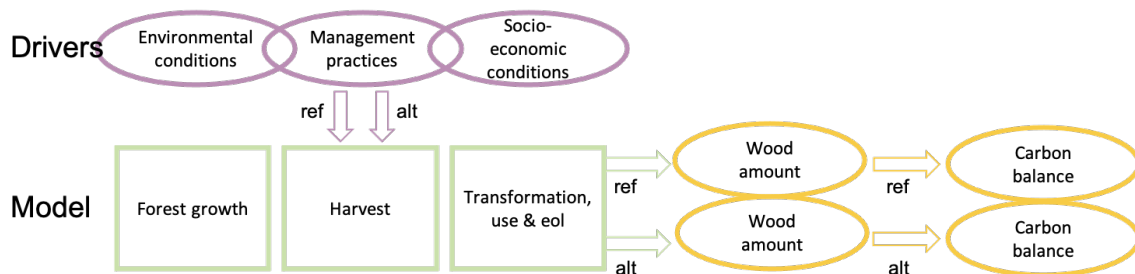


Figure 1: Meta-analysis framework

The steps of the extraction are described sequentially in the present protocol :

- Web of Science query
- Paper selection
- Meta-data extraction
- Data extraction

2 Web of science query

AB=((("wood*product*" OR "timber" OR "wood residues"OR "wood waste" OR HWP OR "wood harvest" OR "wood production") AND ("carbon balance" OR "greenhouse gas*" OR "carbon dioxide"OR "carbon sink" OR "carbon *flux*" OR "CO2 flux*" OR "CO2 emission*" OR "carbon emission*" OR "GHG emission*" OR "climate change mitigation" OR (LCA AND("global warming" OR "radiative forcing"))) AND (scenario* OR substitution OR LCA OR "life cycle" OR "displacement factor*" OR "avoided emission*" OR "GHG saving*" OR "biogenic carbon" OR "biogenic CO2" OR "C* debt") AND ((silviculture OR "rotation" OR "species" OR "clearcut" OR "forest harvest*"OR "management practices" OR "harvest*intensity" OR "harvest*frequency" OR thinning OR "wood harvest*")OR ("supply chain*" OR HWP or "value chain*" OR "end-of-life" OR industr* OR cascad* OR "bioenergy facilit*") OR ("innovat*" OR technolog* OR "low*energy building" OR "high*rise"))))

Between 2000-01-01 and 2022-01-24.

3 Paper selection

Before starting the actual data collection, each contributor will check that each paper fits the inclusion criteria. In case of exclusion the justifying criteria is recorded in the column "Exclusion".

Table 1: Ranked criteria for paper inclusion in the meta-analysis and label to record the exclusion reason in the database

Priority rank	Criteria definition	If criteria is not matched, record in column "Exclusion"
1	The paper is a scientific journal article in English	"no journal "
2	The paper is original research, not a review article	"review"
3	The paper contains estimates of either carbon sequestration or emission . Carbon values can either be absolute (e.g MtC, tCO ₂) or relative to wood, energy production or provided service (e.g kgCO ₂ /MWh, MtC/Mtoe, tC/m ²). Other impact variables are additional variables and their presence is only recorded as a metadata without the extraction of data. If a paper only contains the latter, the paper should not be considered.	"No C output"
4	The C variable is cumulated or can be cumulated over a period of time (possibly infinite in the case of traditional LCAs) .	"C var cannot be cumulated"
5	The carbon sequestration/emission estimates are associated with a change in, or two options for, one of the practice drivers (silviculture, supply chain, technology), resulting in at least two explicit and distinct scenarios Drivers describing changes in surrounding conditions (Environmental change) are recorded and the associated data extracted, but are not sufficient for paper selection (eg. change in climatic conditions) if they cannot be linked to a change in practices as main driver . For example changes in wood demand, prices, ... with a quantified impact on wood harvest are associated with the "unspecified harvest increase" driver. Changes in wood demand, prices, ... with no information on wood harvest may be recorded as "increased forest area" driver if relevant. In both cases, the paper is included.	"no driver"
6	The paper contains estimates of wood amount harvested or used for both carbon balance estimates.	"no wood"
7	At least one C estimate for two compared scenarii rely on identical perimeters. Only those pairs with identical perimeters are extracted.	"different perimeters"
8	If all criteria are met, the paper is selected for data extraction	"included"

After the data extraction phase, a verification phase will take place where all excluded papers will be double-checked by a second reviewer ensuring no paper was erroneously excluded.

Data needed to meet the inclusion criteria can be taken from the article's main body, the supplementary materials or companion technical papers if referred to explicitly in the paper.

4 Database structure

Each line of the spreadsheet records the comparison in the carbon balance due to a change in at least one driver and provides the context of this data comparison. The implication is that one paper can be the object of **several lines if several variables, drivers, model boundaries, or repetitions are covered by the paper**. In such case, the metadata in each line may differ or not. Records may have similar or different values for each column depending on whether they represent repetitions of the same assessment, or different driver evaluations.

Table 2 : The database is organized into 3 levels of sections.

Metadata	Paper description	Authors	
Metadata	Paper description	Journal	
Metadata	Paper description	Publication	
Metadata	Paper description	Topic	
Metadata	Protocol	Exclusion	
Model	Context definition	Location	
Model	Context definition	Sector scale	
Model	Context definition	Time period	
Model	Context definition	Wood affected by driver change	
Model	ProcessesDetails	SubstitutionShortcut	
Model	Processes	C fluxes	Biogenic in-situ C
Model	Processes	C fluxes	Biogenic ex-situ C
Model	Processes	C fluxes	Fossil in-situ
Model	Processes	C fluxes	Fossil ex-situ
Model	Processes	Dynamics	C dynamics
Model	Processes	Dynamics	Market-based feedbacks
Model	Boundaries	Cboundaries	
Model	Boundaries	TimeBoundaries	
Drivers	Change in practices	Silviculture	
Drivers	Change in practices	Supply chain	
Drivers	Change in practices	Technology	
Drivers	Driver	driver description	
Drivers	Environmental change	Demand	
Drivers	Environmental change	Environmental change	
Outcomes	Non_c_outcomes	biogeochemicalOutcomes	
Outcomes	Non_c_outcomes	biophysicalOutcomes	
Outcomes	Non_c_outcomes	ecologicalOutcomes	
Outcomes	Non_c_outcomes	economicalOutcomes	
Outcomes	Non_c_outcomes	LCAOutcomes	
Outcomes	Carbon_outcome	Wood amount	
Outcomes	Carbon_outcome	Cumulated C balance	
Outcomes	Carbon_outcome	Substitution coefficient	

Grey-header columns are automatically calculated columns that should not be manually filled out.

Columns of sections "Metadata" describe the publication details, they are identical for all records of a given publication. All other columns describe the experiments that are extracted from the publication. The paper details are extracted from the Webofscience search tool. Details include source title, author keywords, keywords plus, authors, article title, abstract, journal ISO abbreviation, publication year, DOI. When two papers have the same first author and year, the column multipaper_index is filled with a,b... to differentiate the papers. The column PaperID is automatically generated from the previous columns in the form of author_year or author_yearindex. For example Chen_2020a and Chen_2020b.

Columns of section "Model" record the **model setup to carry out the experiment in terms of geographical scale, time horizon, wood product category represented, C processes included in the model** (and not the whole model capacity if some of the model components are not accounted for in the reported experiments). They are a shared description of both scenarios. If an article's results can be disaggregated **with respect to model boundaries, each model boundary will be recorded in a separate line**. For example in a typical LCA reporting explicitly all steps from material extraction to end-of-life, several pairs of carbon balance will be defined. One record will be defined as only including forestry emissions and manufacturing emissions and the driver, wood and C values will report the results for this boundary set. Another record will be defined as including forestry emissions, manufacturing emissions and maintenance emissions and record the appropriate values etc... The number of boundaries' combinations to be included as studies is left to expert judgment. Different time steps may also be recorded in different lines. For example, when the paper's results show a dynamic trend for C emissions from 2010 to 2100, one study will be described as having time boundaries 2010 to 2050 and

will report driver, wood and C change for this time period. Another record will be described as having time boundaries 2010 to 2100 and will report driver, wood and C changes for this second time period. They will therefore have different time horizons. Expert judgment is used to decide which time periods should be included as studies to report the variability that appears in the results.

Columns of sections “Drivers” describe what distinguishes both scenarios of a recorded pair of carbon balances and assigns it to a category.

Columns of section “Outcomes” record the numerical values for wood and carbon balances of both scenarios and the origin of numerical values in the original publication.

5 Extracting data

5.1 Defining pairs of data

Recorded pairs of data describe the effect on carbon balance of a change in an actual bio, socio or economic strategy, addressing one of the drivers of changes in practices listed in the sheet and in **Table 4**. Changes in environmental drivers are recorded as well given the publication has at least one record for a driver relying on a change in practice (

Table 5).

For all pairs of data (i.e. for each line), 4 types of values will be searched as listed in **Table 3**. The results have to be displayed in terms of at least 2 estimates for carbon sequestration/emission or a difference between 2 estimates. It can cover in-situ C fluxes only, emissions from HWP’s life cycle only, both in-situ and value chain , or for some large scale studies, can encompass emissions from other sectors (transport, agriculture...).

When results are displayed separately for **different system perimeters** (for example, LCA emissions and in-situ biogenic C balance are both displayed explicitly), **both will be recorded in separate lines** and differentiated by filling the associated “C variable perimeter” boxes.

5.1.1 Single vs multiple drivers

In each line, only the driver that explains the recorded change in carbon balance is reflected in the line, and not all drivers evaluated in the paper or addressed by the model. All drivers’ effects on C balance quantified in a paper should be recorded in separate lines. The driver is recorded under two forms, the assignment to one of the pre-identified categories and the explicit names of the two scenarios reflecting the effect of the driver.

When more than two scenarios are displayed for a given driver (for example three levels of intensity), one of the scenarios is defined as the reference and all other values are considered as alternative scenarios and compared to the reference as a separate pair, i.e in a separate line in the table. When many scenarios are displayed in the form of a sensitivity analysis, an expert judgment may be used to only extract some of the data pairs based on results’ redundancy. In particular, when the driver’s effect is roughly linear, only the most extreme scenarios might be extracted.

When extracting the data, different drivers should be recorded on separate lines if their respective effects on the C balance are disentangled in the results shown in the paper. When multiple drivers are identified for a single scenario and their individual effects cannot be disentangled, an expert judgment is made on the **primary driver** that is expected to have the largest effect or from which the other drivers result. This **primary driver is recorded with a 1**, secondary drivers are identified with a 2.

Table 3 : Recording driver values, wood amount and C balance, and expected calculation steps.

Category	Data status	Data extraction procedure	Calculation steps expected
Driver description	Mandatory , numerical or categorical	describe explicitly the driver and give the name of the reference and alternative scenarios as referred to in the paper , for tracking back the recorded data	None required
Non-C outcomes	Optional, binary	Record the non-C variables that are also evaluated in the paper without recording their values	None required
Wood amounts	Mandatory, two options only	Qwood stageInput : Roundwood or HWP. Refers to the wood quantity that is listed in the following section.	None required
	Mandatory, numerical	Values for the wood quantities mobilized associated to both reference and scenario will be recorded in columns 'Qwood_ref' and 'Qwood_alternative' or 'wood_delta' (if the values for each scenario are not available). Wood mobilized may be whole harvest, HWP or recycled wood depending on the C balance calculated. Qwood should refer to the same perimeter as the carbon estimates. For example, if the carbon balance is calculated for the whole harvested wood, then Qwood should include both energy and materials wood products. If the driver is recycling the Qwood Following the guidelines to select the reference, Qwood_alt>Qwood_ref. Qwood_delta is first calculated in the same unit as given in the paper but is then converted into m3 of wood using the named variables listed in the Conversions sheet. QwoodInPublication records the paper's display item from which the wood data have been extracted (for ex. Fig 2, Table 3..)	Qwood_delta = Qwood_alternative – Qwood_ref Conversion : Qwood_delta_m3wood
Cumulated C balance	Mandatory (optional only if a substitution value is provided)	Record the C balance cumulated (or averaged over time). By convention, sequestration in the biomass will be recorded as positive, emissions to the atmosphere will be recorded as negative. For double-check, the sign of the C flux is recorded in the C_sign column. If annual values are provided, compute the cumulated value ensuring the recorded time period in the "Context definition" section is correct. The change in carbon emission/sequestration between reference and scenario will be recorded in priority in the two columns 'C_ref' and 'C_scenario'. When the values for both reference and alternative scenario are not displayed, the C balance might as well be recorded as 'C_delta' if available.	Cratio= (Calternative – Cref) / abs(Cref) Cdelta= Calternative -Cref Conversion : Cdelta_tCO2
Substitution	Mandatory using wood amount and C balance recorded values	Relative carbon balance is calculated, that is the change in C balance divided by the change in wood use. Record the data in category Substitution after checking the correct conversion and ensuring correspondence (same perimeter) of Qwood_delta_m3wood and Cdelta_tCO2. Be specific on the unit (eg. m ² of house floor or m ² of forest instead of m ² , m ³ of harvest or m ³ of wood used as window pane instead of m ³).	Substitution= Cdelta_tCO2 / Qwood_delta_m3wood

Table 4 : Changes in practices' drivers with their definitions and recommendations to define the reference.

Drivers' category	Drivers	Details	Recommendations to define the reference
Silviculture	Unspecified harvest increase	When it is not said by which mechanism the harvest is increased	Scenario with lowest wood use or Scenario with the lowest harvest is the reference
	Area subject to harvest	Effect on a territory's C balance of a change in harvested area	Reference will be scenario with lowest wood use or scenario with lowest managed area
	rotation length	Scenarios with different harvest age	Scenario with lowest wood use or Scenario with the longest rotation length
	cutting intensity	Scenarios test the effect of : whole tree harvest (WTH), thinning or not , clearcutting or not, structural retention, continuous cover forestry, residues removals (slash and stumps) for their effect on in-situ C	Scenario with lowest wood use or most extensive scenario with lowest harvested volume per hectare (eg. no thinning, no WTH, ...)
	species	Compare C balance of stands of different species	Scenario with lowest wood use or Scenario with the highest share of deciduous (deciduous < mixed < coniferous). If not applicable, scenario with the highest species diversity.
	tree density at plantation	Planting density at stand establishment, (different from thinning recorded in cutting intensity)	Scenario with lowest wood use or Scenario with the lowest planting density
	fertilisation	Several levels of fertilization	Scenario with lowest wood use or Scenario with the lowest fertilization
	fertility	Differences in location recorded as a difference in region, climate, type of soil	Scenario with lowest wood use or Scenario with the lowest fertility
Supply chain	harvesting system	Level of mechanization, type of technology used to extract the wood	Scenario with lowest wood use or Scenario with the least efficient or least mechanized technology
	location of industry	Compare two transformation chains based on their location	Scenario with lowest wood use or Scenario with the least productive/efficient location
	structural change in supply chain	change in wood allocation to timber, pulp, energy and on-site residues (ex: debris left in the forest vs used for energy), transportation distance	Scenario with lowest wood use or Scenario with the highest allocation to residues, then energy (if draw on residues), then pulp, scenario with longest transportation distance
	Recycling	Different rates of recycling of manufactured wood products (excluding use of waste from wood mills)	Scenario with lowest wood use or Scenario with the lowest recycling rate
	end-of-life disposal	Incineration, landfilling	Scenario with lowest wood use or Scenario with the highest share of landfilling
	Product lifespan	Time of product use before disposal	Scenario with lowest wood use or longest lifespan
Technology	efficiency improvement	Same technology used but with different efficiencies (including waste share in wood mills and its rate of use as energy/other purposes).	Scenario with lowest wood use or Least efficient scenario
	CCS	Effect of carbon capture and storage on the C balance	Scenario with lowest wood use or Scenario with lowest share of CCS
	technologies/design switch	Example : Conventional versus passive house	Scenario with lowest wood use or Scenario with the most carbon intensive technology

Table 5 : Environmental change drivers with their definitions and recommendations to define the reference.

Drivers' category	Drivers	Details	Recommendations to define the reference
Climate and environment	Water table	Depth of water table	Scenario with least favorable growing conditions
	Temperature	Climate scenarios	
	rainfall	Climate scenarios	
	CO2 fertilization	Ex : FACE experiment	
	disturbance	Fire, area burnt, storms	
	N deposition		
Demand		When policy actions are not quantified as gdp, consumption or price Ex : Deforestation reduction scenario, bioenergy targets	Scenario with lowest wood use or Policy leading to least efficient / least intensive mix of practices (see above)
	Unspecified policy		
	gdp, population, consumption	Scenarios of wood consumption per capita, GDP scenario.	Scenario with lowest wood use or Scenario with lowest wood consumption
	Exogenous wood, energy or material price	Sensitivity to discount rate, material price	Scenario with lowest wood use or Scenario with lowest price competitiveness of wood
	Exogenous C price	Carbon price, carbon credit, shadow price of carbon	Scenario with lowest wood use or Scenario with lowest carbon price / carbon payments
	Exogenous land use change	LUC is defined in exogenous scenarios ex: forestland-to-cropland after logging or post-agricultural forest growth	Scenario with lowest wood use or Least intensive scenario

5.1.2 Defining the reference

For each pair, the reference will be chosen in a systematic way so that the reference is the option with the **lowest value for wood consumption**, or if equal, will be assumed linked to the option with less harvest, less efficient processes, less conifers, less recycling, less fertile and can be chosen following recommendations in **Table 4** of this protocol's document.

When data displayed as deltas, the reference chosen by the authors might not be the same as our convention. In this case the delta is recorded with a minus sign.

5.1.3 Wood and carbon estimates

Each numerical value recorded will be associated with its **variable name, unit, values and source** in the original paper for both reference and alternative scenarios, or difference between scenarios if absolute values are not available. When different units are available for one of the recorded values, an expert judgment will be made to decide which one is recorded.

5.2 Model description data

Model description data describe the model setup corresponding to the extracted pairs of data, not the whole available model. Most model description data items are coded by 0 and 1. Sometimes, expert judgment may be needed if an element is suggested but not clearly mentioned. Only when an item cannot be filled up using expert judgment, the code NE (not explicated) may be used.

5.2.1 Location

The geographical situation of the study is recorded in two columns: 'scale' and 'location'. The scale of the study is coded according to the list below. For scales Europe, country, region and city, in addition, record the country of the study (for Europe, repeat Europe). If two countries are involved in the study, record the upper administrative unit of both countries. For example if supply chains of Sweden and Finland are compared, 'scale' is Eur, 'Location' is Europe. For generic ecosystems the type of ecosystem is recorded as the scale of study (temperate, boreal, tropical forests)

Table 6 : Geographical context definition

Geographical situation of the study	Scale	Location
World	w	0
Temperate forests	tef	0
Tropical forests	trf	0
Boreal forests	bof	0
Europe	eur	Europe
Country	coy	Country or upper common administrative unit if several countries
Region or other subnational administrative unit	reg	Country of the study
City	cit	Country of the study
Single forest stand	std	Country where the stand is located
Individual product	prd	Country of the study

5.2.2 Time period of the study

The time frame of the study is recorded in columns 'time_start', 'time_end' and 'time_length' depending on the type of data available in the publication. If start time (or end time) is expressed as an interval (eg. 2014-2018), use the middle of the interval (here 2016). If there is no indication of time, set time_length to 0. If there is an indication of time length but not of the start and end times, set time_start and time_end to 0 and time_length to the reported value.

5.2.3 Wood sector boundaries considered in the study

The fraction of the wood sector addressed in the recorded comparison is recorded in the section 'Wood sector boundaries'. This section refers to the type of product considered, that is the category of wood product involved in the change in practices or surrounding conditions. Types of products **not affected by the driver change** are noted with 0, types of products affected are noted with 1. "Upstream" means that the study is not focused on a specific type of wood product, but rather on changes in harvest level/practices which may in turn affect the volumes of all product types without this being explicitly described. "Economy_sector" means that the study computes a carbon budget that goes beyond the wood sector (eg. energy production including a mix of feedstocks, wood being one of them but never the only one).

5.3 C pools and fluxes accounted for in the study

The objective of the recording of these metadata is to be able to identify a relationship between boundaries of the model and substitution estimates. All accounted processes are listed, organized by biogenic, and external energy-related.

Table 7 : Process boundaries included in the model

Category	Column name	Detailed name	When to set a 1?
Biogenic C processes	soilC	Soil+litter+deadwood sequestration	Is one of these reservoirs accounted for? Litter refers here to litter from natural processes. Judgment expert may be used based on the understanding of the model used
	Harv_residues	Harvest residues emissions	Covers residues resulting from management practices
	Live_biomass_C	Live biomass sequestration	Covers above and below-ground biomass
	Product_storage_C	Biogenic C storage in the HWP	Is the storage of biogenic C in HWP accounted for?
	Eol_biogenic	HWP end of life C release (emissions)	Is the release of the C content of HWP at the end-of-life accounted for? The alternative is the C neutral assumption of wood products.
	Off_product_biogenic	Biogenic C benefits after product end-of-life due to recycling or avoided emissions	Are the C benefits beyond the product itself taken into account? For example C benefit from recycling or avoided emissions in other sector?
External energy	Forestry_emiss	Forest operations	Which life cycle stages are accounted for with respect to external energy needs and C-intensive processes? Manufacturing_emiss or maintenance_emiss are ticked if such emissions are explicitly accounted for. Where a substitution effect is accounted for (eg. Fossil fuel used for transportation/heating/cement production/... in the reference scenario) these processes are implicitly included and the corresponding items must be ticked as well.
	Manufacturing_emiss	Manufacturing including transport from the forest to the mill, and from the mill to the users	
	Maintenance_emiss	Operation, repair, maintenance	
	Eol_fossil_emiss	End of life fossil including transport to landfill recycling or burning facility	

5.4 Displacement factors

In some case (off_product_biogenic=1 necessary but not sufficient condition) a displacement factor is used to compare carbon balances for the whole or a part of the study. This is recorded in the columns about substitution vision.

Table 8 : Reporting the use of displacement factors

Column name	Details
Self-calculated DF	Is a DF self-calculated? In such a case, the 4 above sections will describe the boundaries underlying the DF.
Displacement factor used	Are external displacement factors used? In such a case, the 4 “Process boundaries” sections will describe the boundaries underlying the DF. If the boundaries underlying the external DF are not described and cannot be “guessed”, previous steps will be NE.
Energy DF origin	If an external displacement factors is used for energy record Author+year
Material DF origin	If an external displacement factors is used for timber record ecord Author+year
Overall DF origin	If an external displacement factors is used for all wood use record ecord Author+year

5.5 Accounting of processes dynamics

The explicit modeling of time dimension both with respect to in-situ sequestration (forest dynamics and biogenic decay varying over time), fossil emission processes (burning/recycling varying over time), market-based mechanisms exogenous (LUC) or endogenous (rebound effects) is recorded in columns biogenic_dyn, fossil_dyn, LUC_dyn, rebound_dyn. Then the eventual use of displacement factors is recorded as well with their references when they are taken from another publication.

Table 9 : Model description of system's dynamics

Time boundaries	Dyn_biogenic	Dynamic processes in-situ	Is the time variability of biogenic C sequestration and emissions account'ed for? An opposite example would be using an average storage rate in soil.
	Dyn_fossil	Dynamic emissions fossil	Is the timing of LCA fossil emissions accounted for or are all emissions assumed to occur at the same time?
	Dyn_LUC	Dynamic land use change	Does the model account for a time-dependent change in land use?
	Dyn_rebound	Dynamic rebound effect	Does the model account for a price-mediated change in system's behaviours?

6 Organizing data extraction

6.1 Extraction traceability

To aid in tracking any issue with data extraction in a given paper, I recommend that for each paper with data, the data extraction process is performed within an additional spreadsheet in the same file where data are recorded for each contributor, easy to find by naming it by the 'PaperID'. This spreadsheet will contain all intermediate data manipulations and calculations (ideally, all documented).

Depending on the data extraction tool used by each data contributor, another possibility would be to create a folder named after 'PaperID' storing:

- Data extraction spreadsheet
- Paper and all published data sources from the paper (supplementary tables, figures).
- Images in individual files, maybe stored in a subfolder, used for programmatic data extraction (e.g. using MetaDigitiser).

6.2 Data digitizing software

In cases where data are not available from tables or supplementary files, we will have to use digitizing software. Some of us with more recent experience with digitizing data from published sources have used WebPlotDigitizer (<https://automeris.io/WebPlotDigitizer/>). WebPlotDigitizer can be used as a web app or installed locally on your PC. It's easy to use and has a lot of functionalities. It seems especially suitable for automatic point detection. Check this tutorial for more information : <https://www.youtube.com/watch?v=P7GbGdMvopU&feature=youtu.be&t=30s>

7 Check list for data quality

After all data extraction is completed, a subset of records will be quality checked by ensuring the confirmation of all the below mentioned items:

- Can the number of records be confirmed?
- Is the driver name consistent with the driver category?
- Can you find the Qwood_ref, Qwood_alt and C_ref, C_alt values in the publication? In the extraction notes?
- Do the Qwood and C have consistent perimeters (for example m3/year, tCO2/year)?

- Is the carbon sign correct?
- Is the unit conversion correct?
- Is the substitution estimate consistent with the paper's conclusion (more wood is beneficial/detrimental)?
- Is the model description consistent with the recorded pairs (and not with the whole model described in the paper)?
- Is the model description consistent with the driver?
- If the model accounts for avoided emissions, is `off_product_biogenic` set to 1?
- If the substitution value is an outlier (DF in column 'Outlier?'), do you understand why ?